Superstore Sales Performance Dashboard (Excel)

Project Overview

This project analyzes a retail **Superstore sales dataset** to evaluate sales performance, identify top-performing products, and understand revenue and profit trends across different regions and product categories.

The analysis was conducted using **Microsoft Excel**, where raw transactional data was cleaned, transformed, and analyzed using **Pivot Tables, Pivot Charts, and an interactive dashboard**.

The objective of this project is to transform raw data into **clear business insights that can support data-driven decision making**.

Dataset Source:
https://www.kaggle.com/datasets/aditirai2607/super-market-dataset

Dataset Description

The dataset contains retail transaction records from a supermarket business.

Key Fields in the Dataset

- **Order Date** – Date when the order was placed
- **Region** – Geographic sales region
- **Category** – Product category (Furniture, Office Supplies, Technology)
- **Product Name** – Name of the product sold
- **Sales** – Revenue generated from each transaction
- **Profit** – Profit generated from each sale

These fields were used to analyze **sales performance, product demand, profitability, and sales trends over time**.

Data Cleaning and Preparation

Before conducting the analysis, the dataset was cleaned and prepared to ensure the data was accurate and structured.

The following steps were performed:

- Checked for **blank or missing values**
- Checked for **duplicate records**
- Converted **Sales and Price columns to Currency format**

- Converted **date columns to proper Date format**
- Converted relevant fields to **Number format**
- Structured the dataset as an **Excel Table** to improve data analysis
- Verified consistency of data types across columns

These steps ensured that the dataset was **clean, organized, and ready for analysis**.

Data Analysis Process

After cleaning the data, analysis was performed using **Excel Pivot Tables and Pivot Charts**.

Several analytical views were created to explore business performance.

1. Sales vs Profit Analysis

A **combined column and line chart** was created to compare:

- Total Sales
- Total Profit

Across different **product categories and regions**.

This helps identify which product categories generate the most revenue and profit.

2. Sales Trend Over Time

A **line chart** was created to analyze **sales trends by order date**.

This visualization helps identify:

- Seasonal sales patterns
- Growth trends
- High performing quarters

3. Top Products by Sales

A Pivot Table was created to identify the **Top 10 products by total sales**.

However, due to dashboard layout and readability considerations, the **Top 5 products were displayed on the dashboard** to highlight the most impactful revenue-generating products.

This approach keeps the dashboard **clear, focused, and easy to interpret**.

Dashboard Design

An interactive **Excel dashboard** was designed to present the analysis clearly and visually.

Key Performance Indicators (KPIs)

- **Total Sales:** $2,326,534.35
- **Total Profit:** $292,296.81

These KPIs provide a quick overview of overall business performance.

Interactive Filters (Slicers)

Users can interact with the dashboard using filters for:

- **Product Category**
- **Region**

These filters allow users to dynamically explore sales performance across different business segments.

Visualizations Included in the Dashboard

The dashboard contains the following elements:

- Sales vs Profit comparison chart
- Sales trend over time chart
- Top performing products table
- KPI summary cards
- Interactive slicers for filtering

Key Business Insights

Technology Category Generates the Highest Revenue

The **Technology category generates the highest sales**, indicating strong demand for technology products.

Businesses should consider prioritizing **inventory and marketing efforts** for this category.

Sales Peak in the Fourth Quarter

Sales consistently **increase during Q4**, indicating strong seasonal demand toward the end of the year.

Increasing **marketing campaigns, promotions, and inventory levels** during this period could increase revenue.

A Small Number of Products Drive a Large Portion of Revenue

The analysis shows that **top-performing products contribute significantly to total sales**, highlighting the importance of focusing on best-selling items.

Tools Used

- Microsoft Excel
- Pivot Tables
- Pivot Charts
- Excel Dashboard
- Data Cleaning Techniques

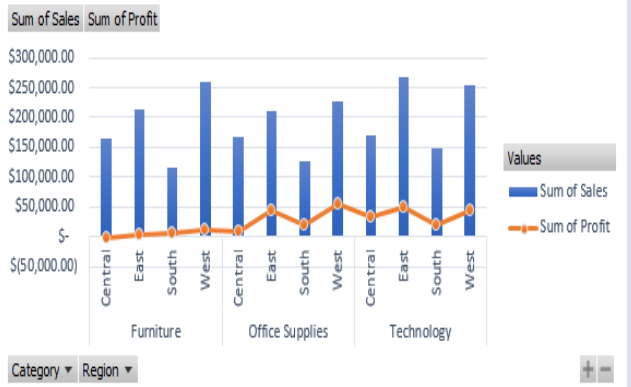
Project Files

The repository contains the following files:

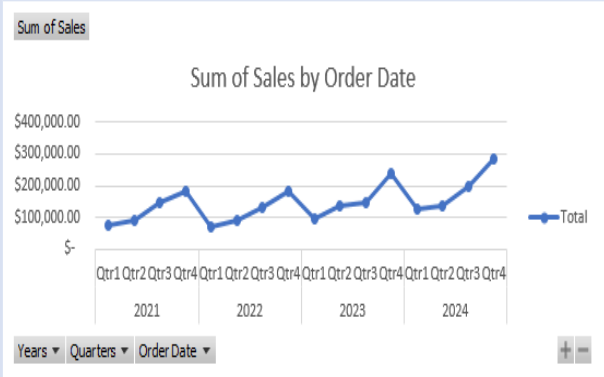
- Superstore Dataset (Excel)
- Cleaned Dataset
- Pivot Tables and Charts
- Excel Dashboard
- Project Report (README)

Dashboard Preview

Superstore Sales Performance Dashboard(2021-2024)



Sum of Sales	Sum of Profit
$2,326,534.35	$ 292,296.81



Category
- Furniture
- Office Supplies
- Technology

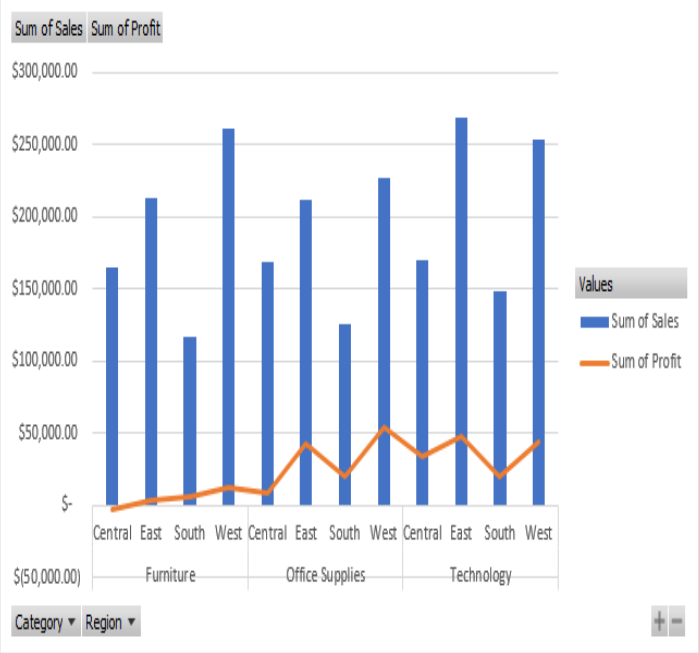
Region
- Central
- East
- South
- West

Top 5 Products	Sum of Sales
Canon imageCLASS 2200 Advanced Copier	$ 61,599.82
Fellowes PB500 Electric Punch Plastic Comb Binding Machine with Manual Bind	$ 27,453.38
Cisco TelePresence System EX90 Videoconferencing Unit	$ 22,638.48
HON 5400 Series Task Chairs for Big and Tall	$ 21,870.58
GBC DocuBind TL300 Electric Binding System	$ 19,823.48
Grand Total	$ 153,385.74

The Top 5 products table is not connected to slicers due to limitations in Excel 2016. This can be resolved using Power Pivot or recreated in Tableau for full interactivity.

Key Insights:
Technology generates the highest revenue, indicating strong customer demand. The business should prioritize inventory and marketing efforts in this category to maximize returns. Sales consistently peak in Q4, suggesting seasonal demand. Increasing promotions and stock availability during this period could significantly boost revenue.

	Sum of Sales	Sum of Profit
Furniture	$ 754,747.76	$ 19,730.00
Central	$ 164,537.65	$ (2,802.21)
East	$ 212,231.70	$ 3,444.74
South	$ 117,298.68	$ 6,771.21
West	$ 260,679.73	$ 12,316.25
Office Supplies	$ 731,893.31	$ 126,023.44
Central	$ 168,216.71	$ 8,970.08
East	$ 211,658.40	$ 42,996.74
South	$ 125,651.31	$ 19,986.39
West	$ 226,366.89	$ 54,070.23
Technology	$ 839,893.28	$ 146,543.38
Central	$ 170,416.31	$ 33,697.43
East	$ 267,938.07	$ 48,441.78
South	$ 148,771.91	$ 19,991.83
West	$ 252,766.99	$ 44,412.34
Grand Total	$ 2,326,534.35	$ 292,296.81



Skills Demonstrated

This project demonstrates key data analysis skills including:

- Data Cleaning
- Data Preparation
- Data Transformation
- Exploratory Data Analysis
- Data Visualization
- Dashboard Development
- Business Insight Generation
- Data Storytelling

Future Improvements

Possible future improvements for this project include:

- Creating a **Tableau or Power BI dashboard version**
- Performing **SQL-based analysis**
- Adding **sales forecasting**
- Conducting **customer segmentation analysis**

Author

Keren Oyem
Junior Data Analyst

GitHub
https://github.com/kerenoyem-hash

LinkedIn
https://www.linkedin.com/in/keren-oyem